Pointers

Lab questions, set 6

51. Assume an integer variable x = 25.
Write a C program using call by reference to increase the value of x by 15 inside a function and display the updated value in main().

52. Assume two integers a = 40 and b = 60.
Write a C program to swap the values of a and b using pointers and a user-defined function.

53. Assume an integer array arr[5] = {5, 10, 15, 20, 25}.
Write a function that receives the array using a pointer and adds 5 to each element. Display the updated array in main().

54. Assume an integer array arr[6] = {45, 12, 89, 23, 7, 34}.
Write a C program to sort the array in ascending order using pointer arithmetic only (no array indexing inside the sorting logic).

55. Write a function that returns a pointer to the variable having the larger value, if two value passed as pass by reference and display the value in main() using dereferencing.

56. Write a C program that uses a function pointer to call a function multiply() which returns the product of two numbers.

57. Write a C program using function pointers to perform the following operations based on user choice:
    i. Addition
    ii. Subtraction
    iii. Multiplication

58. Assume an integer x = 258.
Write a C program to assign the address of x to a char * pointer using type casting and display the value of the first byte stored in memory.

59. Assume an integer variable num = 100.
Write a C program that:
    i. Stores the address of num in a void pointer
    ii. Typecasts the void pointer to an integer pointer

iii. Displays the value of num using the dereferenced pointer

60. Assume the user enters size = 5 and array elements : {11, 22, 33, 44, 55}.
Write a C program that:

i. Dynamically allocates memory using malloc()
ii. Stores the integers using pointers
iii. Displays the elements using pointer arithmetic
iv. Frees the allocated memory